Battery Chemistry and Configuration Review

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Abstract

The battery in an electric vehicle is a critical and diverse component in several different applications. Overall design selection is based on specific requirements for each vehicle, such as: structural significance, weight tolerance, weight placement, total capacity, power requirements, thermal restrictions, cost, and other factors that are analyzed in trade studies. The battery chemistry and configuration will focus on requirements for the battery itself, and exclude the integration of the battery into the PWC. For the application of the Swarm, the series of battery packs in the production PWC, the overall configuration must meet the voltage and amperage requirements for the maximum safe output of a selected motor for each model. Each EEL pack (Echelon Emitting Lithium-Ion) will consist of a fixed number of cells, separated for cost-efficient maintenance in the event of single cell fatigue or failure, with ergonomic handling properties that allow manual removal for transport.

1 Requirements

Tritone aims to target the rental market and compete with existing models. The most prominent considerations in the rental market for PWC selection include lifetime cost and maintenance intervals. To create a marketable selling point, Tritone must meet or exceed the power output, and reduce the lifetime cost and maintenance intervals.

1.1 Power

The typical voltage range for electric vehicles spans between 100V for offroad motorcycles such as the Surron, just over 400V nominal for applications such as the Tesla Model S Plaid, and up to 800V for axial motors such as the YASA YM360. DC power output is modeled as a multiple of amperage and voltage, meaning that higher voltage battery packs will output the same power at a lower draw to the battery. While lower voltage configurations will be lighter and less expensive, it comes at the tradeoff of higher stress on the battery, higher thermal values at operating conditions, and shorter endurance values.

In order to meet the market standard 130hp output of a typical rental ski, such as the Sea-Doo GTI Pro, the battery must be able to provide at least 100kW of power to match the horsepower equivalent. The battery will aim to produce a nominal voltage between 200-400V and discharge an amperage between 250-500A.

1.2 Total Capacity

The average PWC rental lasts approximately an hour, with lower end of rentals lasting 30 minutes and full day rentals lasting 8 hours. For an event such as a day-long tour on a PWC, usage before

docking will vary by location. To simplify future trade studies and advertisement, an assumption of 90-120 minutes between docking will be made. To meet this standard, a new battery pack should have the capacity to operate for a minimum of 90 minutes of standard use.

1.3 Thermal Properties

The packs will be cooled with ambient water serving as a heat exchanger through designated channels. With prolonged full-throttle application, heat will generate in the motors and batteries. As voltage drops with batteries losing charge, power draw will increase, which increases heat generation even further. The selected chemistry shall pose a low risk of thermal runaway using ambient water as a cooling source. Furthermore, the selected chemistry shall have usability in water temperatures that vary from 40 degrees Fahrenheit to 90 degrees Fahrenheit to prevent failure during extreme summer and winter conditions.

During the winter, battery packs should be equipped with thermal measuring systems that allow passive heating in the event of freezing temperatures to protect the longevity of the batteries. High-voltage heaters can be implemented, however an important forewarning to remove batteries and place in a heated environment should be recommended.

1.4 Weight

For the purpose of stability, efficiency, and handling, the weight of a PWC is important to the hull design and overall performance. A lightweight configuration is optimal, targeting a mark that, combined with the electric motor, is lighter than the engine and fuel load from an ICE PWC. Since the electric motor remains lighter than the internal combustion equivalent, the battery load can outweigh a typical fuel load. For the example of the Sea-Doo GTI Pro, the fuel capacity is between 16 and 18 gallons, which gets users approximately 3-5 hours of use in an economy mode. Assuming gas to be between 6 and 7 pounds per gallon, the total fuel load averages out to around 110lbs. For Swarm, the target weight for a battery pack is approximately 300lbs. Tradeoffs on weight and capacity will be evaluated as the pack approaches final design.

1.5 Cost

During the prototyping phase, budget constraints limit the amount of money that can be spent on a proper battery pack. The cell type optimal for production will be used in a smaller scale version for prototyping, limiting the power output of the motor but serving as a baseline for output, performance, and reliability.

In production, the overall cost of the ski shall be optimized to profit off a sale at 17K USD, meaning battery costs will range based on quality and performance, and an optimal pricepoint will be selected based on non-negotiable costs, such as the frame and motor that are selected.

1.6 Safety

Safety of the pack and the user lie within several root causes, each with manageable solutions.

1.6.1 Water Exposure

Batteries will require a suitable Ingress protection rating to meet a sealed factor of safety from a saltwater environment. Furthermore, humidity buildup will need to be mitigated from heat cycling,

requiring a solution that includes condensation control, proper venting, or some form of moisture management.

1.6.2 Corrosion

Saltwater and salty air are extremely corrosive, requiring marine-grade coatings for any exposed or non-exposed metal used to assemble and secure the battery pack.

1.6.3 Grounding

High-voltage battery packs if assembled incorrectly can turn the hull of the vehicle and the surrounding water into a circuit, requiring an isolated DC system for the safety of the user and anyone in the surrounding water.

1.6.4 Thermal and Mechanical limitations

If the PWC is left in the heat or if cooling is insufficient, thermal runaway poses a great risk of spontaneous combustion and is a relatively common failure mode.

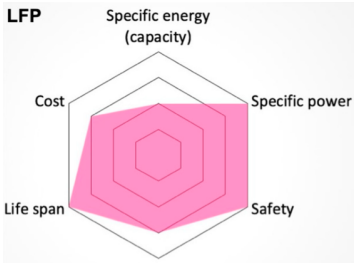
Vibration is also a key consideration for battery packs in a marine application, due to wake and other high impact G forces. While there is evidence for vibrations and shocks to cause inefficiencies in batteries, the studies performed that yielded any results were upwards of 150 G forces, which will never be seen in marine applications without other severe failures.

2 Description and Applications of Chemistries

2.1 LFP

Strengths: Cycle life, Safety

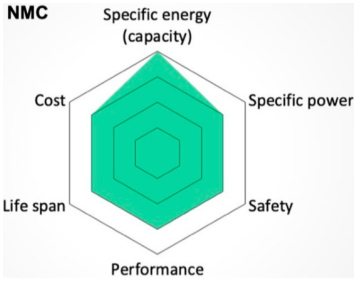
Lithium Iron Phosphate reports unmatched lifespan due to it's high cycle count and thermal stability. The energy and power density is relatively low, and nominal voltage is 3.2V per cell. The most practical application for this chemistry is seen in energy storage, such as solar farms, and short-range EV.



2.2 NMC

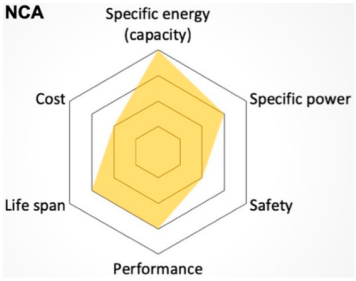
Strengths: Energy density, Availability

Lithium Nickel Manganese Cobalt is a common chemistry for the 18650/21700 cell format. While it has high energy density and a nominal voltage between 3.6 and 3.7V, it is more prone to thermal runaway and requires a thorough BMS. Widely used on e-bikes and light-weight applications



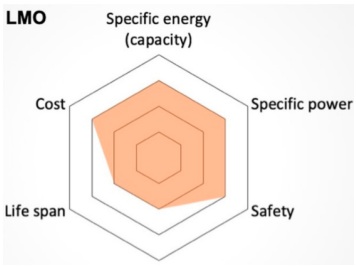
2.3 NCA

Strengths: Extremely high performance, long-lasting, scale applications



Lithium Nickel Cobalt Aluminum has a nominal voltage between 3.6 and 3.7V and is a commonly used chemistry for EV companies with high power due to their incredible energy density and longevity. When kept at proper conditions, it is a strong option for scale batteries. Unfortunately, it is very thermally unstable and requires an absolutely precise BMS. Used commonly by companies like Tesla and Panasonic.

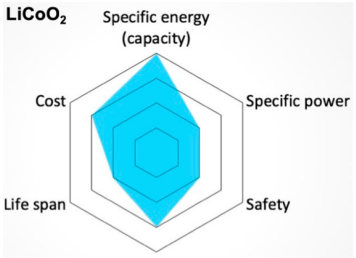
2.4 LMO



Strengths: Discharge, Thermal stability

Lithium Manganese Oxide carries a nominal voltage between 3.7V-3.8V and has great peak burst current. It has better thermal stability than NMC but offers a lower energy density and half of its lifespan. Capacity fades fairly quickly, which is why it is often times blended with other chemistries in battery packs. Most common uses are seen in power tools.

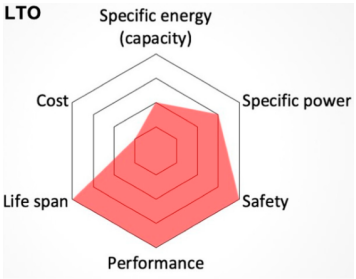
2.5 LCO



Strengths: Energy density

Lithium Cobalt Oxide has a nominal voltage of 3.6V and excels in compact battery packs. Similar to LMO, this chemistry has a short lifespan (slightly higher than LMO), but has much less thermal stability. It has good energy density, but is not stable enough for larger battery packs. Instead, it is used in small lightweight electronics.

2.6 LTO



Strengths: Life cycles, Thermal stability

Lithium Titanate Oxide batteries have a lower nominal voltage of 2.4V per cell, giving them poor energy density figures. However, their safety, recharging speed, and lifespan are excellent. Operating temperatures range typically from -40 C to +60 C. Despite the high life cycle quantity, they also feature incredible discharge rates. The most common application for these batteries are in buses, marine, and heavy machinery.

2.7 Na-Ion

Strengths: Cost, Thermal stability

Sodium ion chemistries are starting to become increasingly more popular due to their potential. Unfortunately, the commercial market for Na-Ion is relatively new, meaning long-term research is not available and availability is limited. This will be a chemistry variant to keep an eye on for several applications, including the Swarm.

3 Trade Studies

A trade study is conducted on the seven different battery compositions, factoring in the requirements from section 1. Each requirement will carry a prescribed weight, indicating its importance to the design.

Table 1: Trade Study Criteria and Weight

Criteria	**Weight**
Discharge	.15
Energy Density	.275
Thermal Properties	.2
Cost	.275
Safety	.1

Each composition is ranked between 1-5 for its performance in each category for the application of a high-voltage and high discharge EV PWC.

Table 2: Trade Study Comparison

Criteria	**LFP**	**NMC**	**NCA**	**LMO**	**LCO**	**LTO**	**NaI**
Discharge	3	4	4	4	1	5	2
Energy Density	3	4	5	2	3	1	2
Thermal Properties	4	3	2	4	1	5	4
Cost	5	4	3	3	2	1	2
Safety	5	3	1	4	1	5	4

4 Results

Performance in each category was multiplied by the weight of the category, and each categorical score was summed to give an overall score of each chemistry out of 5. The highest score will be selected as the chemistry that will be used for the Swarm prototype.

Table 3: Trade Study Results

Chemistry	**Score**
LFP	3.95
NMC	3.7
NCA	3.3
LMO	3.175
LCO	1.825
LTO	2.8
NaI	2.6

With a score of 3.95, LFP is selected as the best battery chemistry for the application of a commercial EV PWC. The cost, safety, and energy properties of LFP gives it an edge over the rest of the competition. NMC, despite having better energy density, narrowly lost out due to higher cost and lower thermal stability.

NCA and LMO are both great power sources. Holding strengths in different categories, the NCA battery is unstable with very high energy density, and LMO is stable with low energy density. Neither chemistry is cost efficient.

LTO is an excellent chemistry for its discharge, safety, and longevity. In the application of a PWC, these factors are not as important as cost and energy density. Similarly, Na-Ion suffers in the same categories, but increasing research and development of this chemistry could lead to promising results in these areas.

LCO is the least suitable chemistry for large-scale packs. It features low discharge, horrid thermal stability, and a disproportionate cost due to short lifespan.

5 Conclusion

The battery chemistry trade study indicates that Lithium Iron Phosphate (LFP) is the most suitable chemistry for the Tritone Swarm PWC prototype. The configuration of the battery cells to reach the target power demands would be 89s2p, featuring 178 pouch cells in total.

The analysis also establishes several important design targets. The battery system should be capable of supplying approximately 100 kW of peak electrical power, and the most efficient battery layout to achieve this power goal is selected to save weight.